

Beliefs about Refugee Integration in Germany: A Survey Experiment on Ukrainian and Syrian Refugees

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Supervisor: Prof. Dr. Florian Zimmermann

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Lidiia Levytska
Matriculation Number: 50274891

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1 Introduction

2 Literature Review and Conceptual Framework

2.1 Immigration Attitudes and Migrant Characteristics

A large literature examines how host-country populations evaluate migrants and which characteristics shape these evaluations. One central distinction in this literature is based on whether attitudes toward migrants primarily reflect individuals' own economic interests and labor-market considerations or characteristics of the migrants.

Earlier economic studies provide some evidence that labor-market considerations are associated with immigration preferences (e.g., Scheve and Slaughter, 2001; Mayda, 2006). Beyond economic self-interest, immigration attitudes have also been linked to broader fiscal, cultural, and group-related concerns (Burns and Gimpel, 2000; Dustmann and Preston, 2007; Sides and Citrin, 2007). Reviewing this literature, Hainmueller and Hopkins (2014) conclude that evidence for explanations based on individual economic self-interest is generally limited.

Experimental evidence further challenges a simple labor-market competition explanation and rather points to the importance of migrants' own characteristics. Hainmueller and Hiscox (2010) find that both high- and low-skilled natives prefer high-skilled over low-skilled immigrants. If immigration attitudes were primarily driven by competition with similarly skilled workers, preferences would be expected to differ according to respondents' own skill levels. Instead, the common preference for high-skilled immigration suggests that respondents place considerable weight on migrants' human capital and potential economic contribution. This pattern is also found in multidimensional experiments. Higher education, occupational status, employability, and language proficiency are generally associated with more favorable evaluations of migrants (Hainmueller and Hopkins, 2015; Bansak et al., 2016; Czymara and Schmidt-Catran, 2016). More recent evidence from Germany similarly shows greater willingness to accept skilled economic migrants with recognized professional qualifications, good language skills, and a definite job offer (Wolff et al., 2025). Taken together, these studies show that characteristics associated with migrants' expected labor-market integration play an important role in how they are evaluated.

Country of origin, cultural and religious background, and migration status also influence attitudes when other characteristics are held constant. Hainmueller and Hopkins (2015) find differences in evaluations across immigrants' countries of origin alongside effects of education, occupation, and English proficiency. In the European context, Bansak et al. (2016) show that preferences over asylum seekers depend jointly on country of origin, religion, employability, and vulnerability. Similar patterns emerge in Germany. Czymara and Schmidt-Catran (2016) find that acceptance varies with migrants' country of origin and reason for migration as well as their education and German-language proficiency.

These studies show that evaluations depend not only on migrants’ individual characteristics, but also on broader group-related characteristics and migration categories.

Although much of this literature focuses on attitudes and preferences toward migrants rather than beliefs about their expected outcomes, it shows that evaluations vary systematically with migrants’ observable characteristics. In line with this literature, higher human capital and stronger language proficiency may similarly serve as signals when respondents form expectations about migrants’ economic and social integration. This leads to the first hypotheses:

H1a: Holding other migrant characteristics constant, respondents expect migrants with higher human capital to achieve more favorable labor-market integration outcomes than migrants with lower human capital.

H1b: Holding other migrant characteristics constant, respondents expect migrants with higher German-language proficiency to achieve more favorable labor-market and social-integration outcomes than migrants with lower German-language proficiency.

2.2 Beliefs and Expectations about Migrant Outcomes

A growing literature shows that individuals hold systematic beliefs about immigrants and their economic and social characteristics. Much of this research examines perceptions of immigrant populations as a whole, including their size, composition, education, and employment (Sides and Citrin, 2007; Alesina et al., 2023). Respondents tend to overestimate the share of immigrants living in their country, and perceive immigrants as less educated, more unemployed, and more dependent on government transfers than they actually are. However, these perceptions may depend on which types of migrants respondents have in mind. Blinder (2015) shows that public understandings of the term “immigrant” do not necessarily correspond to the full migrant population, with some migrant categories being considerably more salient than others. This suggests that treating migrants as a single category may obscure differences in how distinct types of migrants are perceived.

One particularly relevant distinction concerns migrants’ reasons for migration. Evaluations vary between refugees and migrants arriving for economic reasons (Czymara and Schmidt-Catran, 2016), while refugees also tend to experience less favorable actual labor-market outcomes than economic migrants, including lower employment, occupational, and earnings outcomes (Brell et al., 2020; Fasani et al., 2024). Although this evidence concerns evaluations or actual outcomes rather than beliefs about future integration, it provides reason to expect that host-country populations may also distinguish between otherwise comparable refugees and economic migrants when forming such expectations. This leads to the following hypothesis:

H2: Holding all other migrant characteristics constant, respondents expect refugees to achieve less favorable labor-market integration outcomes than economic migrants.

A related literature examines migrants’ employability and labor-market experiences and shows that employment outcomes depend not only on migrants’ individual characteristics, but also on organizational and institutional factors in the host country (Shirmohammadi et al., 2019; Farashah et al., 2023). Research on immigrant recruitment further demonstrates that employers’ perceptions can shape migrants’ employment opportunities and that migrant workers may face discrimination and disadvantages in hiring and occupational outcomes (Almeida et al., 2015; Dietz et al., 2015). However, this literature primarily examines migrants’ actual labor-market experiences or employers’ recruitment decisions rather than expectations held by the broader host-country population.

A smaller literature comes closer to examining such expectations from the perspective of the host-country population. Niemann et al. (2024), for example, study how working adults in Germany form expectations about migrants’ occupational future and integration efforts depending on migrant characteristics such as origin, skill level, and reason for migration. Related research demonstrates that individuals also hold beliefs about discrimination. Haaland and Roth (2023) document substantial heterogeneity in beliefs about racial discrimination in hiring in the United States. Although their study concerns racial discrimination rather than migration, it demonstrates the conceptual distinction between beliefs about discrimination and actual discriminatory behavior.

Taken together, existing research provides evidence on several individual dimensions of migrants’ expected integration, but evidence examining expectations across multiple economic and social outcomes remains limited. This thesis contributes to this literature by examining expectations concerning employment, occupational match, income, discrimination, and social integration for Ukrainian and Syrian refugee migrants and economic migrants in Germany.

2.3 Differential Perceptions of Ukrainian and Syrian Migrants

Earlier literature shows that responses to refugees in Germany already differed across groups and changed over time during and after the 2015–2016 refugee inflow. In particular, country of origin was associated with differences in acceptance of refugees, while other characteristics such as migration circumstances and family composition also shaped public responses (Bansak et al., 2016; Czymara and Schmidt-Catran, 2016, 2017; Plener et al., 2017; Liebe et al., 2018). Czymara and Schmidt-Catran (2017), for example, show that following the events of New Year’s Eve 2015/2016, the acceptance of refugees generally increased, whereas the acceptance of immigrants of North African and Arab origin declined. Other research demonstrates that attitudes also varied with local exposure to the refugee inflow. Albarosa and Elsner (2023), for instance, document heterogeneous changes in attitudes across German regions depending on the size of the local refugee inflow. Looking at a later period, Przechomski et al. (2026) find that, after an initial increase in negative sentiment, attitudes gradually became more positive between 2017 and 2021.

The arrival of Ukrainian refugees following Russia’s invasion of Ukraine in 2022 provided a new context for examining differences in responses to refugee groups in Europe. Existing research demonstrates that Ukrainian refugees have generally received more favorable public and institutional responses than Syrian refugees (Moise et al., 2024; Drewski and Gerhards, 2025; Das, 2025). Evidence from Germany similarly shows greater willingness to help Ukrainian refugees, more positive perceptions of their warmth and integration prospects, and greater solidarity toward them than toward Syrian refugees (Echterhoff et al., 2023; Schmidt et al., 2024; Stefaniak et al., 2025). These findings therefore suggest that public responses are shaped not only by refugee status itself but also by refugees’ country of origin.

Several explanations have been proposed for these differences. Theories of group position and perceived group threat suggest that attitudes toward immigrant groups depend partly on how these groups are positioned and perceived relative to the host population. Building on the group-position perspective of Blumer (1958), Quillian (1995) argues that prejudice can reflect perceived collective threat and the relative position of the in-group and out-group. In the context of Ukrainian and Syrian migrants, existing studies similarly point to differences in perceived cultural similarity, threat, and deservingness (Echterhoff et al., 2023; Schmidt et al., 2024; Stefaniak et al., 2025). At the broader political level, Drewski and Gerhards (2025) show that Syrian and Ukrainian refugees were framed differently in Germany and Poland and link these differences to cultural and geopolitical factors that shaped state responses to the two groups.

These findings provide reason to expect that these differences between Ukrainian and Syrian migrants may also be reflected in beliefs about their integration outcomes. In line with this literature, I develop the following hypothesis:

H3a: Holding other migrant characteristics constant, respondents expect Ukrainian migrants to achieve more favorable labor-market and social integration outcomes than Syrian migrants.

In addition, the previous discussion in Section 2.2 provides reason to expect that individuals also form beliefs about the discrimination in labor-market outcomes of migrants. Combined with the less favorable perceptions of Syrian relative to Ukrainian migrants documented in this section, I developed the following hypothesis:

H3b: Holding other migrant characteristics constant, respondents expect Syrian migrants to face greater discrimination in the German labor market than Ukrainian migrants.

Further, I am interested in whether the difference between Ukrainian and Syrian migrants may depend on their reason for migration. Much of the literature above discussing differential responses toward the two groups concerns refugees specifically. Much of the literature above discussing differential responses toward the two groups concerns refugees specifically, suggesting that country of origin may be particularly consequential when migrants are evaluated in a refugee context. By contrast, evaluations of economic migrants

may place relatively greater weight on characteristics related to their expected economic contribution. I therefore develop the following hypothesis:

H3c: The difference in expected integration outcomes between Ukrainian and Syrian migrants is greater when migrants are presented as refugees than when they are presented as economic migrants.

In addition, I explore whether differences between Ukrainian and Syrian migrants vary with their level of human capital, and in particular whether higher human capital reduces the country-of-origin gap in expected integration outcomes.

2.4 Belief Formation Through Contact, Experience, Information, and Memory

Intergroup contact is one potential channel through which individuals form and update perceptions of migrant groups. The contact hypothesis, originating with Allport (1954), proposes that contact between members of different groups can reduce prejudice under optimal contact conditions. Among a large subsequent literature, as one major meta-analysis testing this hypothesis, Pettigrew and Tropp (2006) find that intergroup contact is generally associated with lower prejudice but mainly that these effects are not limited to situations in which all of Allport’s favorable conditions are satisfied. Evidence from migration and refugee contexts similarly indicates that direct contact can be associated with more favorable evaluations of migrants and refugees (Kotzur et al., 2019; De Coninck et al., 2021; Khalil and Naumann, 2022).

However, the amount of contact alone may not be sufficient to explain how intergroup experiences shape perceptions. Contact can generate different experiences, and its effects may therefore depend on the nature of the interactions that occur. It is therefore important to explicitly distinguish between positive and negative intergroup contact. Laurence and Bentley (2018) show that greater ethnic diversity can generate opportunities for both positive and negative intergroup contact, where positive contact is associated with more favorable attitudes and perceptions of migrants and refugees, while negative contact is associated with less favorable attitudes and behavioral tendencies (Kotzur et al., 2019; Bagci et al., 2023). Similarly, De Coninck et al. (2021) distinguish the frequency and valence of intergroup contact in the context of the European refugee crisis and find that the valence of direct contact is particularly important for attitudes toward refugees. Evidence from Germany also suggests that contact with foreigners can reduce immigration worries, while repeated contact over time does not necessarily strengthen this relationship (Khalil and Naumann, 2022). Experimental and quasi-experimental evidence from other countries shows that increased exposure to refugees and individuals with a migration background can under some circumstances generate more exclusionary or hostile responses (Enos, 2019; Hangartner et al., 2019). These findings suggest that greater exposure to a migrant

group should not automatically be interpreted as more favorable contact and motivate distinguishing the frequency of contact from the valence of the experiences it produces.

The importance of contact valence is consistent with a broader literature showing that personal social experiences can shape beliefs and behavior. Schwerter and Zimmermann (2020) experimentally show that positive and negative social experiences lead to substantial differences in subsequent willingness to trust, with beliefs about others' trustworthiness playing an important role in these effects. Applied to the migration context, this suggests that personal experiences with members of a migrant group may be associated with subsequent beliefs and expectations about the group more broadly. These findings provide a reason for considering not only whether respondents have interacted with Ukrainian and Syrian migrants, but also the negative versus positive experiences they have had with members of the respective migrant groups.

Individuals may also form evaluations based on the information about migration that they acquire beyond their own direct personal experiences. Experimental evidence shows that providing information about immigrants and refugees can affect subsequent perceptions and attitudes, although the direction and magnitude of these effects depend on the information provided and respondents' prior beliefs (Günther, 2023; Hayo and Neumeier, 2023; Dylong and Uebelmesser, 2026). Dylong and Uebelmesser (2026), for example, find that providing accurate information about the size of the immigrant population in Eastern Germany leads to more supportive attitudes toward immigration, with stronger effects among respondents with less prior contact with immigrants. In contrast, Guenther (2024) finds that correcting respondents' underestimation of the number of asylum seekers increases opposition to immigration. Taken together, these findings suggest that external information represents an additional source from which individuals may form perceptions of migrant groups, and that its influence may be particularly relevant when direct personal experience is limited.

Media constitute an important source of information about migrant groups. The literature on mediated intergroup contact conceptualizes mass media as a form of indirect contact through which individuals can acquire information about out-groups even in the absence of direct personal interaction (Schiappa et al., 2005; De Coninck et al., 2021). Importantly, however, the relationship between media exposure and attitudes depends not only on the amount of exposure but also on the content and framing of the information encountered. Media representations of immigrants and refugees frequently emphasize economic, cultural, or security-related threats (Atwell Seate and Mastro, 2016), while different types of media may portray refugees in systematically different ways (Jacobs et al., 2016). Previous research suggests that direct and mediated contact may interact, with positive direct contact potentially reducing individuals' sensitivity to negative media representations (Ahmed, 2017). Moreover, media exposure may itself be selective, as individuals can choose to consume refugee-related media content that is based partly on their existing views about refugees (Lueders et al., 2019). Taken together, this literature

establishes media exposure as an important indirect source of information about migrant groups and suggests that both the content of this information and individuals' prior direct experiences may be relevant for how migrants are subsequently perceived and evaluated.

However, past experiences and information may not receive equal weight in memory when individuals form current beliefs. The influence of past experiences and information on current beliefs may depend on what individuals recall when forming an evaluation. Models of associative memory suggest that recall is selective and cue-dependent, such that experiences related to the current context are more likely to come to mind and influence beliefs (Bordalo et al., 2023; Enke et al., 2024). Consistent with this mechanism, Graeber et al. (2024) show that stories remain more influential on beliefs over time than statistical information because they are more likely to be recalled. Applied to the migration context, evaluations of Ukrainian and Syrian migrants may therefore depend not only on respondents' overall exposure to each group, but also on which personal experiences, stories, and other information about the respective group come to mind when forming their beliefs.

Taken together, this literature suggests that beliefs about migrant groups may reflect not only the information individuals have encountered, but also how this information was acquired and which experiences are subsequently recalled. This may be particularly relevant for comparisons between Ukrainian and Syrian migrants in Germany, given differences in the timing and context of the two groups' large-scale arrival and, consequently, in respondents' opportunities for direct contact and exposure to group-related information. At the same time, greater exposure to a group does not necessarily imply more favorable experiences or beliefs. The present study therefore complements the experimental comparison of otherwise identical Ukrainian and Syrian migrant profiles with respondent-level measures of group-specific contact and experience, allowing these factors to be examined as correlates of differences in expected integration outcomes.

3 Experimental Design

The study employs a factorial vignette experiment in which respondents evaluate hypothetical migrants whose characteristics are independently varied across profiles. This approach is particularly suitable for the present research question because migrants differ simultaneously along several characteristics, such as country of origin, migration status, skill level, and language proficiency. Factorial designs allow the effects of these characteristics to be distinguished by independently varying them while holding the remaining characteristics constant. The proposed design builds on previous conjoint and factorial vignette experiments examining how migrant characteristics affect evaluations by host-country populations (Hainmueller and Hopkins, 2015; Bansak et al., 2016; Czymara and Schmidt-Catran, 2016).

- i indexes respondents.

- j indexes vignette profiles.
- $g \in \{\text{Ukraine, Syria}\}$ indexes migrant groups.
- Y_{ij} denotes respondent i 's evaluation of vignette j .
- C_{ig} denotes respondent i 's contact exposure to migrant group g .
- Q_{ig} denotes respondent i 's overall contact quality with migrant group g .
- M_{ig} denotes respondent i 's media exposure regarding migrant group g .

General Notation: For inline equations, use For displayed equations, use

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or the equation environment.

3.1 Target Population and Proposed Sample

3.2 Factorial Vignette Design

The study employs a factorial vignette experiment in which respondents evaluate hypothetical migrants whose characteristics are independently varied across profiles. This approach builds on previous factorial and conjoint experiments examining how characteristics of immigrants and refugees affect evaluations by host-country populations (e.g., Hainmueller & Hopkins, 2015; Bansak et al., 2016; Czymara & Schmidt-Catran, 2016).

The vignettes describe potential migrants who currently live in Germany and differ randomly across several characteristics. The description of immigrants is based on a total of five dimensions: country of origin (Ukrainian/Syrian), reason for immigration (war refugee/labor migrant), gender (male/female), language skills (low/good German), and education (low/high). Country and language skills describe the cultural distance. Distinguish between two levels: (1) low and (2) good German language skills. The education factor represents the human capital of the immigrants. Distinguish between a (1) low and a (2) high level of education, where: low level of education: completed secondary education but has no vocational or university degree. high level of education: has a university degree. It should also be taken into account that language skills also have an economic component. Success in the labor market depends, for example, on the ability to speak the local language. The effect size of language proficiency could exceed the effect sizes of religious affiliation and country of origin. However, it should be considered that part of the effect of language skills, in the sense of the human capital hypothesis, can also be attributed to economic considerations. (Czymara & Schmidt-Catran, 2016)

Why religious affiliation is not included in the vignette: As discussed in the literature review, previous research demonstrates that religion and perceived cultural distance can

influence attitudes toward immigrants (e.g., Czymara & Schmidt-Catran, 2016). However, in this study, religious affiliation is deliberately not included as a randomized vignette attribute. The present study focuses on differences in expectations associated with migrants' country of origin, migration status, human capital, gender, and German-language proficiency. Introducing religion as an additional vignette dimension would increase the complexity of the experimental design. Moreover, explicitly providing information about religious affiliation could alter the beliefs that respondents would otherwise associate with Ukrainian and Syrian migrants based on the country of origin alone. Religion is therefore left unspecified in the vignette, allowing the country-of-origin treatment to capture respondents' overall beliefs associated with the respective national-origin group. Consequently, the estimated country-of-origin effect should be interpreted as a composite effect that may encompass perceived cultural and religious differences, rather than as an effect of nationality that is independent of religion. Respondents' own religious affiliation is nevertheless collected as a background characteristic and can be included as a control or examined in heterogeneity analyses.

Vignette wording: [Gender: Mr./Mrs.] [random letter]. came from [country of origin: Ukraine/Syria] to Germany. [He/She] came to Germany as [reason for immigration: a refugee after fleeing the war in [his/her] country / an economic migrant in search of better employment and economic opportunities in Germany]. [He/She] [skill level: has completed a university degree / has no formal vocational qualification], speaks German [language skills: well / only to a limited extent] and is currently looking for stable employment.

vignette universe: $2 \times 2 \times 2 \times 2 \times 2 = 32$ profiles

male/female $\times$ Ukraine/Syria $\times$ refugee/labor migrant $\times$ high/low skill $\times$ good/limited German = 16 profiles

Each respondent receives only a few randomly selected vignettes. The randomization allows estimation of the marginal effect of each characteristic.

1. Explained vignette attributes: *Origin_j MigrationStatus_j Gender_j Education_j GermanSkills_j*

2. Choosing particular vignette attributes: Educational attainment is included because previous vignette and conjoint evidence consistently documents more favorable evaluations of highly skilled immigrants (Hainmueller & Hiscox, 2010; Hainmueller & Hopkins, 2015; Czymara & Schmidt-Catran, 2016). Language proficiency can be connected to Czymara & Schmidt-Catran and other relevant experiments. Migration status can be justified through the refugee-versus-labor-migrant literature. Country of origin obviously comes from your substantive Ukrainian–Syrian research question.

3.3 Outcomes Measures

3.4 Respondent Questionnaire

Citing measurement sources for the respondent questionnaire: C_{ig}^{Work} workplace contact frequency C_{ig}^{Social} social contact frequency Q_{ig} contact quality

Contact quality: The notation $Q_{ig} \in \{1, \dots, 7\}$ denotes respondent i 's overall contact quality with migrant group g . Contact quality is measured on a 7-point scale, where 1 indicates very negative contact experiences and 7 indicates very positive contact experiences. The measure is adapted to distinguish between the two origin groups examined in the present study.

$$PositiveExperience_{ig} = \max(Q_{ig} - 4, 0)$$

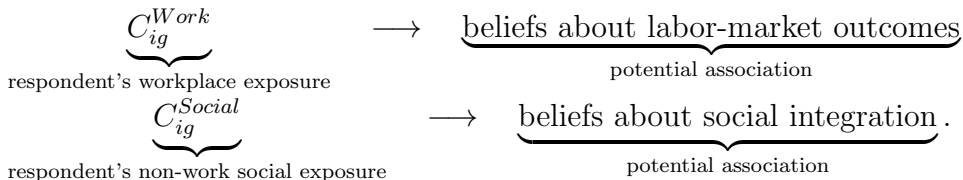
$$NegativeExperience_{ig} = \max(4 - Q_{ig}, 0)$$

Following the intergroup-contact literature, the questionnaire distinguishes between the frequency and quality of respondents' contact with members of the two origin groups. This distinction is motivated by previous evidence showing that the quantity and valence of intergroup contact constitute conceptually distinct dimensions and may have different relationships with attitudes toward refugees (e.g., De Coninck et al., 2021). Contact is measured separately for Ukrainians and Syrians and across workplace and non-work social settings.

Media exposure: $MediaExposure_{ig}$ Media exposure is measured separately for Ukrainian and Syrian refugees. Building on work conceptualizing refugee-related media exposure as a form of indirect intergroup contact (e.g., De Coninck et al., 2021), respondents report the frequency with which they encounter relevant news and media coverage. The measure is adapted to distinguish between the two origin groups examined in the present study.

Motivated by research showing that the memorability of information and narratives may influence subsequent beliefs (Graeber et al., 2024), the questionnaire additionally elicits whether respondents can recall a particularly salient personal experience involving members of either origin group and, if so, the valence of that experience.

Belief about migrant integration:



Beliefs about large communities and network effects:

Large community $\rightarrow$ More information/referrals $\rightarrow$ Better employment
versus

Large community $\rightarrow$ Less native interaction $\rightarrow$ Slower integration.

Construct:

$$\Delta B_i^{\text{Network}} = B_{i,S}^{\text{Network}} - B_{i,U}^{\text{Network}}$$

3.5 Randomization and Survey Procedure

4 Empirical Strategy and Identification

Identification and equations: Baseline equation:

$$Y_{ij} = \alpha + \beta_1 Ukr_j + \beta_2 Ref_j + \beta_3 Female_j + \beta_4 HighSkill_j + \beta_5 German_j + \varepsilon_{ij}.$$

Adjusted specification:

$$Y_{ij} = \alpha + \beta_1 Ukr_j + \beta_2 Ref_j + \beta_3 Female_j + \beta_4 HighSkill_j + \beta_5 German_j + \boldsymbol{\delta}' \mathbf{X}_i + \varepsilon_{ij}.$$

H4 regression with interaction term:

$$Y_{ij} = \alpha + \beta_1 Ukraine_j + \beta_2 Refugee_j + \beta_3 (Ukraine_j \times Refugee_j) + \boldsymbol{\gamma}' \mathbf{Z}_j + \varepsilon_{ij}.$$

where

$$\mathbf{Z}_j = \begin{pmatrix} Female_j \\ HighSkill_j \\ GoodGerman_j \end{pmatrix}$$

$$\mathbf{X}_i = (Age_i, Gender_i, Edu_i, Emp_i, Inc_i, Mig_i, East_i, Religion_i, Right_i)'$$

Then,

$$Y_{ij} = \alpha + \boldsymbol{\beta}' \mathbf{Z}_j + \boldsymbol{\theta}' \mathbf{X}_i + \varepsilon_{ij},$$

where $\boldsymbol{\theta}$ is the vector of coefficients associated with the respondent controls, and $\mathbf{Z}_j$ is contains the randomized vignette characteristics and $\mathbf{X}_i$ contains respondent-level controls.

5 Limitations and Extensions

5.1 Extensions

5.2 Limitations and Identification Constraints

6 Conclusion

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Statement of Authorship

I hereby confirm that the work presented has been performed and interpreted solely by myself except for where I explicitly identified the contrary. I assure that this work has not been presented in any other form for the fulfillment of any other degree or qualification. Ideas taken from other works in letter and in spirit are identified in every single case.

Date:

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